COLLEGE MANAGEMENT SYSTEM USING SQL

Managing student records, course information and fee details.

Name: Devadharshini S

Course: DA

| Project | 2026

PROJECT INTRODUCTION

- This project focuses on managing and analyzing college data using SQL.
- The aim of this project is to organize and store information about students, teachers, courses, enrollments, and fees in a structured relational database.
- The project involves:
  - → Building a relational database with multiple tables.
  - → Writing SQL queries for data retrieval and analysis.
  - → Managing student records and course enrollments efficient.
  - → Extracting insights related to student data and course management.

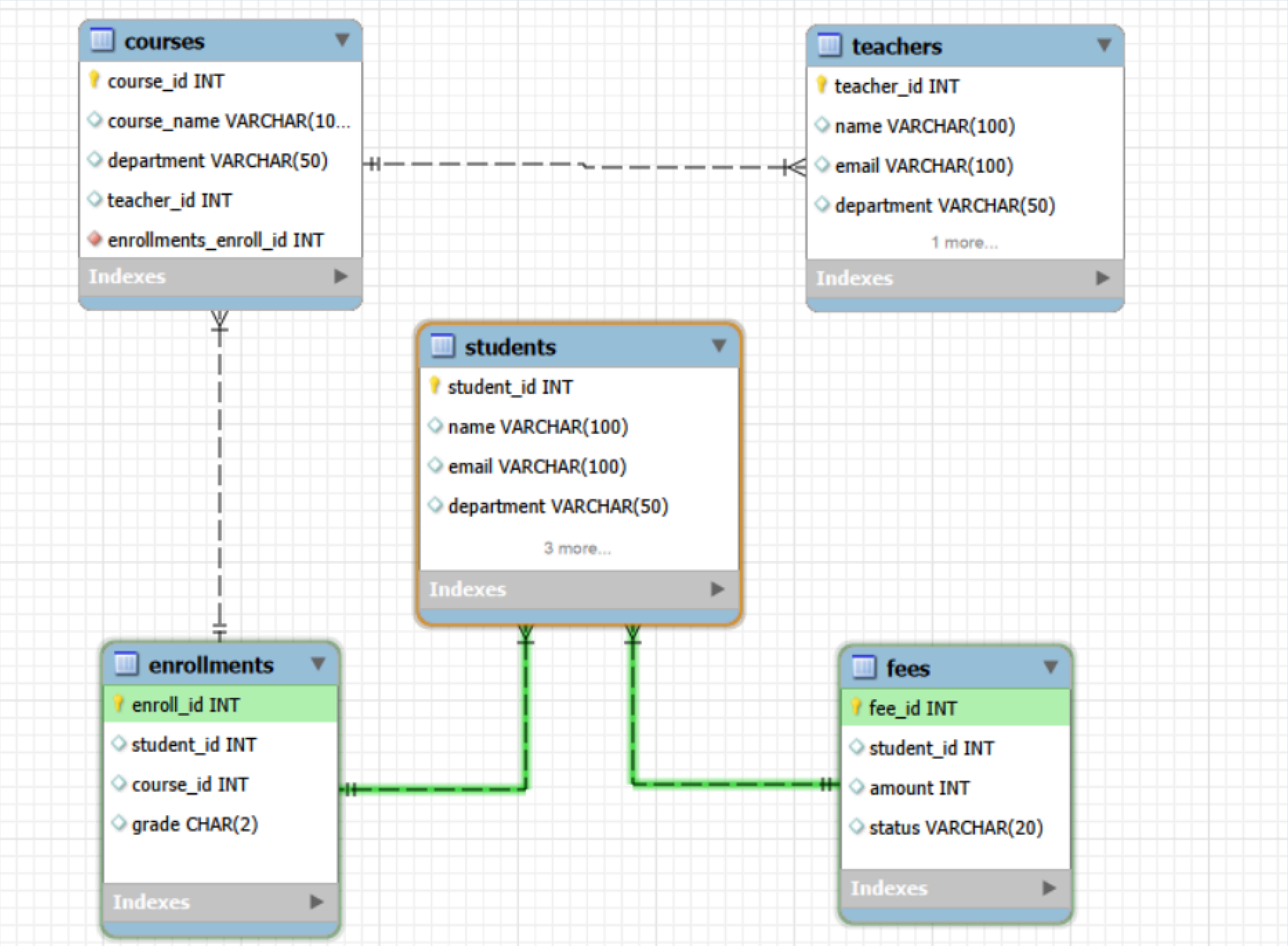
PROBLEM STATEMENT

- Colleges generate a large amount of data such as Student information , Teacher details , Course records , Enrollment data , Fee payment information .
- The challenge is to:
- Organize this data into a structured relational database.
- Manage student-course relationships efficiently.
- Retrieve useful information using SQL queries.
- Maintain accurate and organized academic records.

PROJECT OBJECTIVE

- Build a relational database for college management.
- Store student, teacher, course, and fee information.
- Write SQL queries from basic to advanced level.
- Analyze data using filtering, grouping, and joins.
- Demonstrate real-world database management using SQL.

DATABASE SCHEMA (ER DIAGRAM)



DATASET INFORMATION

Students –

student_id	name	email	department	year
1	Yolanda Harris	longjared@gmail.com	IT	1
2	Heather Ray	lmurray@cobb.com	ECE	2
3	Nicholas Burnett	terrellchristopher@gmail.com	IT	4
4	Hannah Wise	seth27@chavez.biz	EEE	3
5	Rachel Hughes	phillipssteven@wolf.com	IT	3
6	Wesley Brown	nixondavid@hotmail.com	EEE	4
7	Matthew Aguirre	csharp@phillips.com	IT	3
8	Amy Johnson	smithedward@walker.info	IT	4
9	Dorothy Stephenson	jschwartz@gmail.com	IT	3
10	Ruben Alvarez	vsullivan@yahoo.com	ECE	2
11	Joshua Gutierrez	zweber@yahoo.com	IT	3
12	Louis Giles	frankrivera@yahoo.com	IT	2
13	Sylvia Elliott	cindy61@barnett.com	CS	4
14	Melanie Stewart	simmonsandrew@hicks.com	EEE	2
15	Adam Chase	derek49@hotmail.com	IT	4

Teachers –

teacher_id	name	email	department
1	Phillip Osborne	greencindy@clark.net	CS
2	Tommy Wolfe	kennethrogers@hotmail.com	ECE
3	Noah Smith	taylorgrant@harris.com	ECE
4	Linda Santiago	melissa77@wilkinson.net	CS
5	Sarah Arnold	jburton@yahoo.com	EEE
6	Sandra Parker	simpsonmanuel@daugherty.com	IT
7	Leah Garcia DDS	simpsonrandy@gmail.com	ECE
8	Mr. Paul Hall	carrillodaniel@warren.info	ECE
9	Michael Sanchez	samuelestrada@gmail.com	EEE
10	Austin Becker	goodkayla@yahoo.com	EEE
11	Carmen Sanchez	cole90@porter-gill.com	EEE
12	Meghan Barnes	jonesleonard@pope.com	CS
13	Jamie Oneill	cgardner@griffin.com	EEE
14	Diana Castillo	lauren47@garcia-roth.com	CS

teachers 1 ×

DATASET INFORMATION

Courses –

course_id	course_name	department	teacher_id
1	Course1	ECE	35
2	Course2	IT	46
3	Course3	CS	79
4	Course4	IT	3
5	Course5	IT	96
6	Course6	IT	49
7	Course7	IT	30
8	Course8	IT	87
9	Course9	CS	9
10	Course10	ECE	21
11	Course11	EEE	83
12	Course12	EEE	87
13	Course13	CS	78
14	Course14	EEE	75

Enrollment –

enroll_id	student_id	course_id	grade
1	175	40	B
2	159	70	C
3	190	40	D
4	42	78	C
5	66	43	B
6	25	43	D
7	120	17	B
8	93	20	D
9	135	11	D
10	46	68	D
11	192	91	B
12	125	96	B
13	28	56	B
14	30	43	A

Fees -

fee_id	student_id	amount	status
1	172	45990	Pending
2	158	29291	Paid
3	126	66077	Pending
4	98	47717	Pending
5	106	60783	Paid
6	128	49494	Pending
7	51	39104	Pending
8	110	27444	Pending
9	152	50997	Pending
10	146	42573	Paid
11	199	25322	Paid
12	176	46058	Paid

QUERIES CLASSIFICATION

- Simple Queries: Basic SELECT, WHERE, ORDER BY
- Intermediate Queries: GROUP BY, HAVING, Aggregates
- Advanced Queries: Joins, Aggregations

SIMPLE QUERIES

1. This query retrieves all student records stored in the database.

```sql
SELECT * FROM Students;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

student_id	name	email	department	year
1	Yolanda Harris	longjared@gmail.com	IT	1
2	Heather Ray	lmurray@cobb.com	ECE	2
3	Nicholas Burnett	terrellchristopher@gmail.com	IT	4
4	Hannah Wise	seth27@chavez.biz	EEE	3
5	Rachel Hughes	phillipssteven@wolf.com	IT	3
6	Wesley Brown	nixondavid@hotmail.com	EEE	4
7	Matthew Aguirre	csharp@phillips.com	IT	3
8	Amy Johnson	smithedward@walker.info	IT	4
9	Dorothy Stephenson	jschwartz@gmail.com	IT	3
10	Ruben Alvarez	vsullivan@yahoo.com	ECE	2

Students 1

SIMPLE QUERIES

2. This query filters students belonging to the CS department.

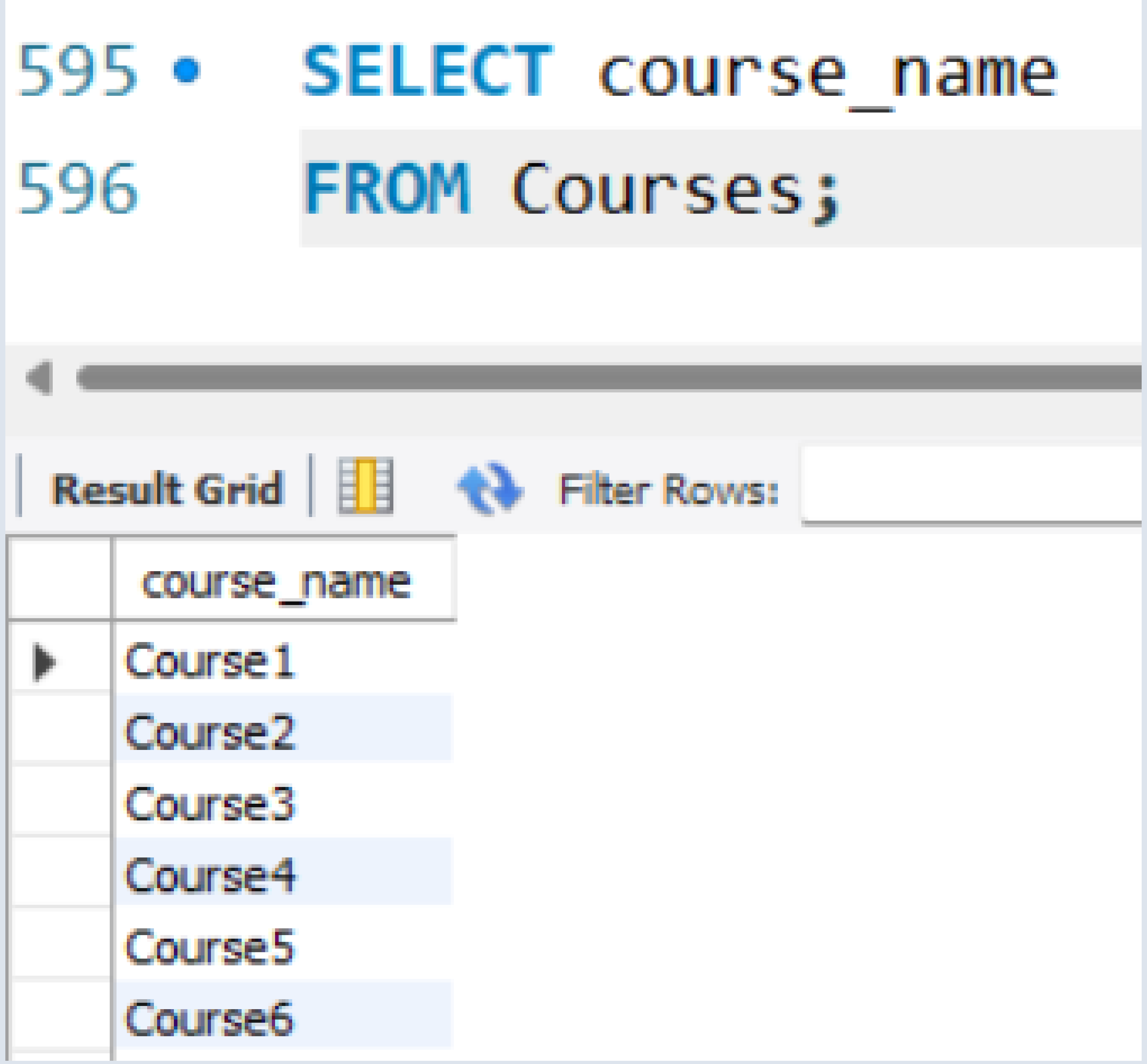
```sql
SELECT * FROM Students
WHERE department = 'CS';
```

Result Grid | Filter Rows: | Edit: | Export/Import:

student_id	name	email	department	year
13	Sylvia Elliott	cindy61@barnett.com	CS	4
21	Vincent Greene	hortondorothy@parker-reed.com	CS	2
22	Fernando Roberts	robert07@yahoo.com	CS	4
25	James Huynh	parksmario@hotmail.com	CS	2
28	Christopher Lee	colleenanderson@yahoo.com	CS	2
32	Mark Middleton	jenkinsdaniel@gmail.com	CS	4
33	Jennifer Payne	justin46@gmail.com	CS	3
34	Derek Black	cordovaolivia@lopez-stone.com	CS	4
35	Mr. Dennis Phillips DDS	goodwinjennifer@rose.com	CS	4
38	Justin West	xscott@robinson.biz	CS	4

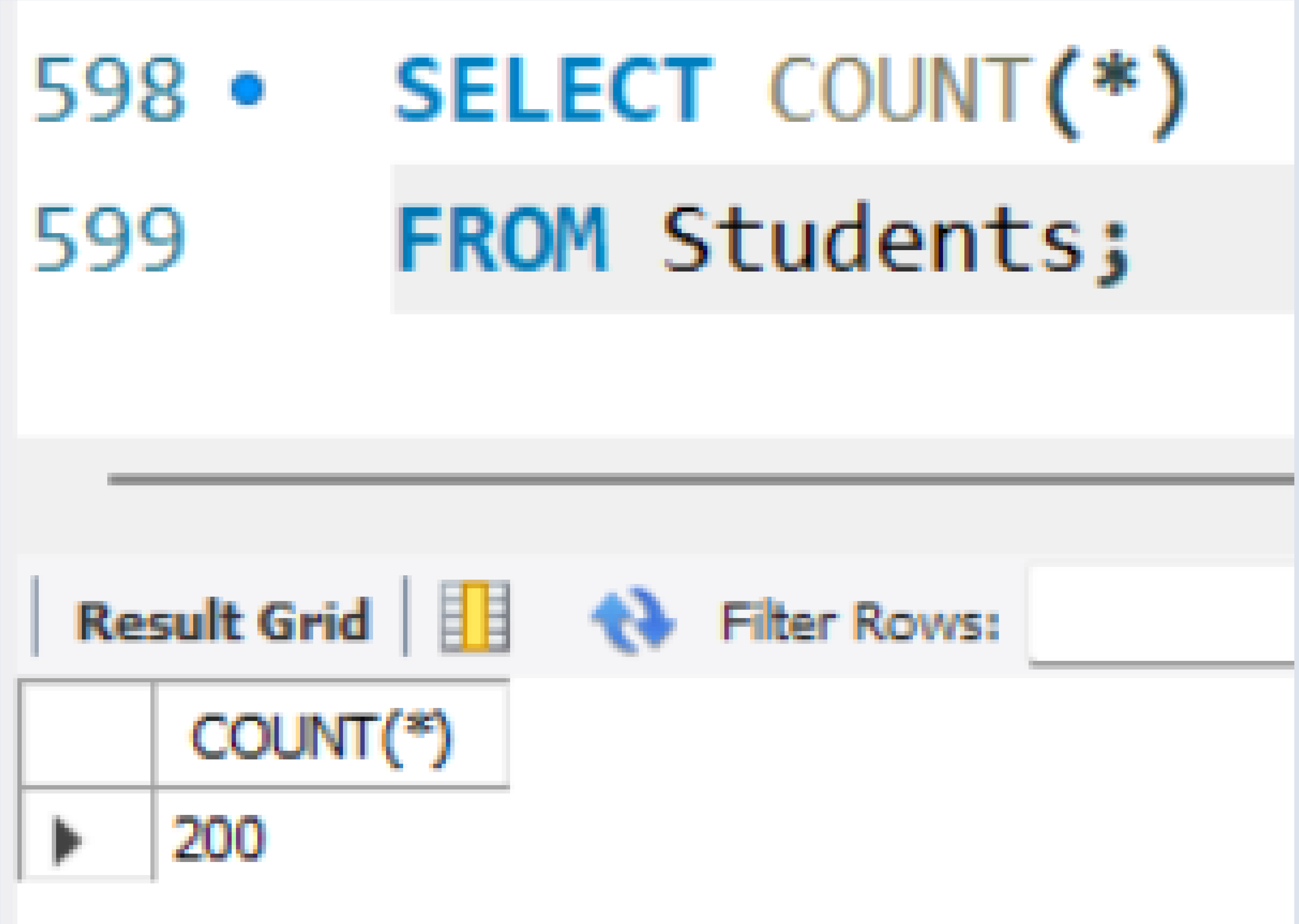
SIMPLE QUERIES

3. This query displays the list of available courses.



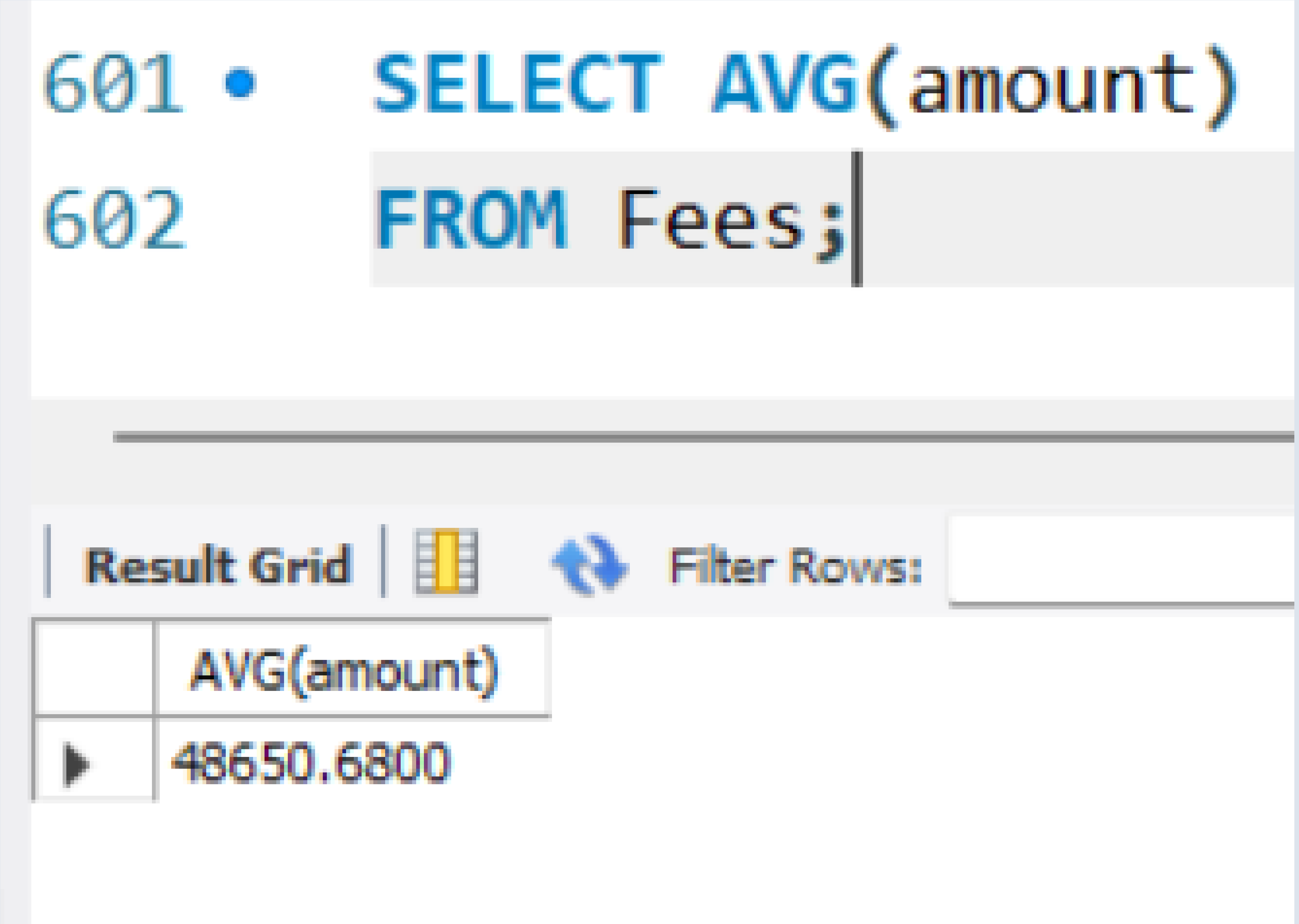
INTERMEDIATE QUERIES

1. This query returns the total number of students in the college database.



INTERMEDIATE QUERIES

2. This query calculates the average fee amount paid by students.



INTERMEDIATE QUERIES

3. This query groups students by department.

```sql
SELECT department, COUNT(*)
FROM Students
GROUP BY department;
```

Result Grid | Filter Rows: | Export:

department	COUNT(*)
IT	67
ECE	48
EEE	35
CS	50

ADVANCED QUERY

1. This query joins three tables to display student-course relationships.

```sql
SELECT Students.name, Courses.course_name
FROM Students
JOIN Enrollments
ON Students.student_id = Enrollments.student_id
JOIN Courses
ON Courses.course_id = Enrollments.course_id;
```

Result Grid | Filter Rows: | Export: | Wrap C

name	course_name
David Murray	Course40
Mr. Bryan Contreras	Course70
Gary Fuller	Course40
Amanda Rasmussen	Course78
Elizabeth Sandoval	Course43
James Huynh	Course43

ADVANCED QUERY

2. This query shows students who completed their fee payment.

```sql
SELECT Students.name, Fees.status
FROM Students
JOIN Fees
ON Students.student_id = Fees.student_id
WHERE status = 'Paid';
```

Result Grid | Filter Rows: | Export:

name	status
Christine Martinez	Paid
Tammy Payne	Paid
Emily Gibbs	Paid
James Johnson	Paid
Brian Short MD	Paid
Joseph Jordan	Paid

ADVANCED QUERY

3. This query shows which teacher teaches which course by joining the Teachers and Courses tables.

```sql
SELECT teachers.name AS teacher_name,
       courses.course_name
FROM teachers
JOIN courses
ON teachers.teacher_id = courses.teacher_id;
```

Result Grid | Filter Rows: | Export: | Wr

teacher_name	course_name
David Brown	Course1
Veronica Pugh	Course2
Carol Richard	Course3
Noah Smith	Course4
Douglas Gonzalez	Course5
Tricia Smith	Course6
Laura James	Course7

KEY INSIGHTS

- Student data can be organized efficiently using relational databases.
- SQL queries help retrieve academic information quickly.
- Course enrollment data helps track student participation.
- Fee records help monitor payment status.
- SQL provides powerful tools for data analysis and

PROJECT BENEFITS

- Improved SQL querying skills.
- Understanding relational database design.
- Hands-on experience with realistic datasets.
- Practical knowledge of database management systems.

CONCLUSION

- The College Management System using SQL helps manage academic data effectively.
- The project demonstrates how relational databases can store and maintain student, teacher, course, enrollment, and fee information in a structured way.
- SQL queries were used to retrieve, filter, and analyze data from multiple tables.

THANK YOU !